

PROVISIONAL PATENT SPECIFICATION

Method and System for Geometric Constraint Surface Diagnostics with Recursive Self-Audit and Cross-Domain Equivalence Detection

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1 TECHNICAL FIELD

This invention relates to methods and systems for diagnostic analysis of engineered, physical, and mathematical systems using the intrinsic geometry of constraint surfaces. More particularly, the invention provides methods for: (a) computing parameterisation-independent geometric invariants of the constraint surface defined by a system's coupling equation; (b) recursively applying the diagnostic to its own outputs to extract hidden structure and produce a self-auditing closure; (c) systematically validating the diagnostic configuration by testing all possible variable-role assignments; and (d) comparing constraint surface geometry across different systems to detect structural equivalence, classify system types, and transfer diagnostic knowledge from characterised to uncharacterised systems.

The invention extends and enhances the diagnostic method described in Australian Provisional Patent Application No. AU 2026902758 (the "parent application"), filed 28 March 2026, by the same applicant.

2 BACKGROUND

2.1 Limitation of Existing Coupling Diagnostics

The parent application describes a method for fault diagnosis in engineered systems by modelling the system as a three-component Directive-Based Product (DBP) triple, computing transpositions of the coupling equation, and evaluating invariant closure across transposition frames. The coupling kernel described therein comprises a spectral channel, a holonomy channel, and a nonlinearity parameter.

While effective, this coupling kernel has three limitations that the present invention addresses:

(a) **Parameterisation dependence.** The spectral and holonomy channels depend on how the system variables are measured and scaled. Different parameterisations of the same physical system can produce different diagnostic values, potentially leading to inconsistent diagnoses.

(b) **Single-pass limitation.** The diagnostic evaluates the system at a single level. It does not examine the structure of its own diagnostic outputs, and therefore cannot detect hidden complexity that emerges only upon recursive self-examination.

(c) **Single-system limitation.** The diagnostic analyses one system at a time. It provides no method for comparing the diagnostic profiles of two different systems to determine whether they share structural features, belong to the same structural class, or exhibit equivalent coupling geometry.

2.2 Limitation of Linear Coupling Assumption

The parent application specifies bilinearity as a formal property of the coupling operation. However, many practically important systems -- including semiconductor junctions, chemical reactions, fluid dynamics, and phase-transition phenomena -- exhibit nonlinear couplings that violate strict bilinearity. The present invention generalises the coupling property to encompass both linear and nonlinear regimes within a unified framework.

3 SUMMARY OF THE INVENTION

The present invention provides four methods that extend the diagnostic framework of the parent application:

Method 1 -- Intrinsic Geometric Diagnostics: A method for computing parameterisation-independent geometric invariants of the constraint surface defined by a system's coupling equation, including the Gaussian curvature, mean curvature, and isotropy ratio. These invariants detect structural features of the coupling -- including critical points, phase transitions, regime boundaries, and coupling symmetries -- that are invisible to spectral and holonomy analysis.

Method 2 -- Recursive Self-Audit (Closure Diagnostic): A method for recursively applying the diagnostic to its own outputs, producing a convergent closure tuple that characterises the system at all levels of self-examination. The recursive procedure is guaranteed to converge within a finite number of iterations by a dimensional cascade, and extracts hidden structure not visible to single-pass analysis.

Method 3 -- Systematic Configuration Validation (Centrifuge): A method for testing all possible assignments of system variables to the Directive, Substrate, and Product roles, identifying the natural coupling configuration, detecting degenerate configurations, and determining the coupling complexity class.

Method 4 -- Cross-Domain Equivalence Detection: A method for comparing the constraint surface geometry of two or more systems to detect structural equivalence, comprising normalised profile comparison, optimal monotonic reparameterisation via quantile matching, destructive interference testing, and reparameterisation equivalence measurement.

The invention further provides a generalised coupling property (Structured Coupling) that encompasses both bilinear and smooth nonlinear couplings, with automatic regime determination via the nonlinearity parameter.

4 DETAILED DESCRIPTION

4.1 Generalised Coupling Property: Structured Coupling

The coupling operation [direct sum] relating the three components of the system triple through D [direct sum] $S = P$ satisfies the following generalised property:

Structured Coupling: The coupling is a smooth (continuously differentiable) function $F(D, S, P) = 0$ relating the three components. In the linear regime, the coupling is bilinear: for fixed Directive, the map from Substrate to Product is linear, and vice versa. In the nonlinear regime, the coupling is a smooth nonlinear function of all three components. The departure from bilinearity is quantified by the nonlinearity parameter κ , defined as the norm of the difference between the coupling and its best linear approximation at the operating point. The diagnostic method automatically determines the appropriate analysis regime based on the computed value of κ : when κ is below a predetermined threshold, the system is diagnosed using linear methods; when κ exceeds the threshold, nonlinear methods are engaged.

This generalised property encompasses the bilinear coupling of the parent application as a special case ($\kappa = 0$) while extending the framework to nonlinear systems including semiconductor junctions, fluid dynamics, chemical kinetics, and critical phenomena.

4.2 Method 1: Intrinsic Geometric Diagnostics

4.2.1 The Constraint Surface

Given the coupling equation $F(D, S, P) = 0$, the set of all (D, S, P) triples satisfying this equation defines a two-dimensional surface (the constraint surface) embedded in three-dimensional variable space. This surface encodes the complete coupling relationship between the three components.

4.2.2 Gaussian Curvature

The Gaussian curvature K_G of the constraint surface at a point is computed as:

$$K_G = \det(BH) / |\text{grad}F|^4$$

where BH is the 4x4 bordered Hessian matrix:

$$BH = \begin{vmatrix} 0 & F_D & F_S & F_P \\ F_D & F_DD & F_DS & F_DP \\ F_S & F_SD & F_SS & F_SP \\ F_P & F_PD & F_PS & F_PP \end{vmatrix}$$

and $|\text{grad}F|^2 = F_D^2 + F_S^2 + F_P^2$ is the squared gradient magnitude.

The subscripts denote partial derivatives of the coupling equation F with respect to the system variables. These derivatives are computed from the coupling equation and the sensor data at the operating point.

The Gaussian curvature K_G is parameterisation-independent by Gauss's Theorema Egregium: it depends only on the intrinsic geometry of the constraint surface, not on how the system variables are measured, scaled, or parameterised. This means that K_G produces identical diagnostic values regardless of the choice of units, coordinate system, or variable scaling applied to the sensor data.

4.2.3 Mean Curvature

The mean curvature H of the constraint surface at a point is computed as:

$$H = (|\text{grad}F|^2 \cdot \text{trace}(H_F) - \text{grad}F^T \cdot H_F \cdot \text{grad}F) / (2 \cdot |\text{grad}F|^3)$$

where H_F is the 3x3 Hessian matrix of second partial derivatives of F.

4.2.4 Isotropy Ratio

The isotropy ratio R is computed from the unit normal vector $n = \text{grad}F / |\text{grad}F|$ as:

$$R = 1 - \min(n_i^2) / \max(n_i^2)$$

where n_i are the components of the unit normal vector.

The isotropy ratio measures how symmetrically the three system components participate in the coupling at the operating point: - R near zero indicates all three components participate equally (symmetric coupling); - R near one indicates one component is effectively decoupled from the other two (asymmetric coupling, indicating a "linear passenger" that enters the coupling equation linearly while the other two enter nonlinearly).

4.2.5 Diagnostic Interpretation

The computed geometric invariants provide fault classification and system characterisation as follows:

(a) **Gaussian curvature K_G :** - K_G near zero: nearly flat constraint surface, indicating quasi-linear coupling consistent with normal operation in many systems; - K_G large and positive: bowl-shaped geometry, indicating strongly convergent coupling; - K_G large and negative: saddle geometry, indicating the coupling curves in opposite directions along different axes -- a structurally significant operating point such as a critical point, phase transition, or regime boundary; - Progressive changes in K_G over time: indicate changes in coupling geometry that may precede acute faults.

(b) **Coupling complexity classification:** - Class 1: the determinant of the Hessian matrix H_F equals zero, indicating at least one component enters the coupling linearly (a "linear passenger"); - Class 2: the determinant of H_F is nonzero, indicating all three components enter the coupling nonlinearly.

(c) **Isotropy ratio R :** - R near zero with high K_G : all components equally involved in a strongly nonlinear coupling; - R near one: identifies which component is the linear passenger, guiding diagnostic interpretation.

4.2.6 Diagnostic Field Extension

The Gaussian curvature K_G may be computed not only at the current operating point but across a range of operating conditions, producing a curvature profile (diagnostic field) that maps the geometric structure of the coupling across its operating domain. This profile is characterised by:

- The critical value λ_1 : the K_G value at any critical point of the coupling;
- The approach exponent α : the rate at which K_G approaches its critical value near a critical point;
- The concentration ratio ρ : the ratio of K_G at the critical point to K_G far from criticality;
- The baseline curvature λ_{∞} : the K_G value far from any critical point;
- The profile shape: classified as Plateau (broadly elevated near critical point), Spike (sharply peaked), Flat (uniformly low), or Regime-dependent (varying across the operating range).

These profile characteristics provide classification capabilities beyond single-point diagnosis, enabling identification of system type, operating regime, and proximity to critical transitions.

4.3 Method 2: Recursive Self-Audit (Closure Diagnostic)

4.3.1 Principle

The diagnostic outputs -- comprising K_G , the operating point parameters, and the coupling-specific profile characteristics -- are themselves quantities that form a three-component relationship. The diagnostic method may therefore be applied recursively to its own outputs, producing a meta-diagnostic that audits the consistency and completeness of the primary diagnostic.

4.3.2 Procedure

The recursive self-audit comprises:

Level 0: Compute the primary diagnostic (K_G , H , R , and profile characteristics) on the original constraint surface.

Level n (n greater than or equal to 1): Identify three quantities from the Level $n-1$ outputs as a new triple. Construct the constraint relating these three quantities. Compute K_G on this meta-constraint surface. This produces the Level n diagnostic value.

Termination: The recursion terminates when $|K_G^n - K_G^{(n-1)}|$ falls below a specified tolerance, indicating no new non-trivial geometric content is being extracted.

4.3.3 Convergence Guarantee

The recursive procedure converges within a finite number of iterations (typically 3 to 4 for three-variable systems) due to a dimensional cascade: each recursive level reduces the dimensionality of the diagnostic output by one degree, progressing from a two-dimensional surface field (infinitely many values) to a one-dimensional profile (a function of one parameter) to a single number to a constant. Once the output is a constant, further recursion produces no new information.

4.3.4 Closure Tuple

The recursive self-audit produces a closure tuple:

$(K_G^0, K_G^1, n^*, K_{G\infty})$

where: - K_G^0 is the primary (Level 0) Gaussian curvature; - K_G^1 is the meta-level (Level 1) Gaussian curvature; - n^* is the closure depth (number of iterations to convergence); - $K_{G\infty}$ is the fixed-point curvature value at convergence.

This closure tuple provides a complete characterisation of the coupling at all levels of diagnostic self-examination. Two couplings with identical closure tuples are diagnostically equivalent at every level.

4.3.5 Class Transition Detection

The recursive self-audit may reveal that the coupling complexity class changes between levels. For example, a Class 1 coupling at Level 0 (linear passenger present) may produce a Class 2 meta-coupling at Level 1 (no linear passenger). This class transition indicates hidden structural complexity not visible to single-pass analysis.

4.4 Method 3: Systematic Configuration Validation (Centrifuge)

4.4.1 Principle

For a system with three measured quantities (x_1 , x_2 , x_3), there exist six possible assignments of these quantities to the Directive, Substrate, and Product roles. The centrifuge diagnostic tests all six assignments to identify the optimal configuration and detect structural properties of the coupling.

4.4.2 Procedure

For each of the six role assignments:

- (a) Compute the forward and inverse transpositions as described in the parent application.
- (b) Compute the coupling kernel (including K_G , K_{spec} , K_{hol} , and κ).
- (c) If the recursive self-audit (Method 2) is employed, compute the closure tuple.
- (d) Record whether the transpositions are computable (the inverse exists and is unique) or degenerate (the inverse does not exist or is not unique).

4.4.3 Outputs

The centrifuge produces:

- (a) **The natural coupling:** the role assignment producing the lowest holonomy residue, the most stable recursive closure, and/or the most geometrically informative K_G values. This is the assignment most consistent with the system's inherent structure.
- (b) **The degeneracy pattern:** which role assignments are degenerate and which are well-defined. The number and pattern of degenerate configurations characterise the coupling complexity: - 0 degenerate: fully invertible coupling (rare); - 2–4 degenerate: typical for nonlinear couplings; - 5–6 degenerate: highly constrained or over-determined coupling.
- (c) **The coupling complexity class:** determined from the Hessian determinant as described in Method 1.

(d) **Resolution limits:** if the coupling kernel values (particularly κ) collapse to zero for specific variable assignments, this indicates a decomposition at which the geometric content vanishes -- a structural limit of the diagnostic at that configuration.

4.5 Method 4: Cross-Domain Equivalence Detection

4.5.1 Principle

Two systems with different physical implementations may share the same underlying coupling geometry. The cross-domain equivalence method detects such shared structure by comparing the geometric profiles of the two systems' constraint surfaces.

4.5.2 Profile Extraction

For each system, compute the K_G profile as a function of a common parameterisation variable (for example, distance from a critical point, operating condition, or normalised input level). Normalise each profile to the range $[0, 1]$ by dividing by the maximum value.

4.5.3 Direct Comparison (Destructive Interference)

Subtract one normalised profile from the other to produce a residual signal. Compute:

- (a) The root-mean-square (RMS) residual: a scalar measure of overall profile mismatch.
- (b) The spectral decomposition (via Fourier transform) of the residual: identifies whether differences are in overall shape (low-frequency components) or fine structure (high-frequency components).
- (c) The geometric content of the residual: compute the curvature of the residual trajectory to determine whether the differences have geometric structure or are parametric only.

4.5.4 Harmonisation via Reparameterisation

If the direct comparison reveals significant mismatch:

- (a) **Power-law reparameterisation:** fit a power-law mapping $\delta_1 = a \cdot \delta_2^b$ between the two systems' parameterisation variables. The parameters (a, b) characterise the rate difference between the systems.
- (b) **Optimal monotonic reparameterisation (quantile matching):** for each normalised profile value v in $[0, 1]$, find the parameterisation values δ_1 and δ_2 at which the respective profiles equal v . The resulting mapping $\delta_1(\delta_2)$ is the optimal monotonic reparameterisation. Apply this mapping and recompute the residual.

(c) **Reparameterisation equivalence:** if the post-harmonisation residual is below a threshold (for example, $\text{RMS} < 10^{-3}$), the two systems' profiles are reparameterisation-equivalent -- they are the same geometric shape at different rates. In differential geometry, two curves related by monotonic reparameterisation are the same curve.

4.5.5 Equivalence Classification

The cross-domain comparison classifies system pairs as:

- **Geometrically identical:** post-harmonisation RMS approximately equal to 0 and residual curvature approximately equal to 0. The systems share the same constraint surface geometry.
- **Parametrically different:** significant pre-harmonisation mismatch but negligible post-harmonisation mismatch. Same shape, different rate.
- **Structurally different:** significant mismatch persists after harmonisation. Different coupling geometry.

4.5.6 Applications

The cross-domain equivalence method enables:

- (a) **Diagnostic transfer:** if a well-characterised system is geometrically equivalent to an uncharacterised system, diagnostic knowledge (fault signatures, operating regimes, critical thresholds) can be transferred from the known to the unknown system.
- (b) **System classification:** systems can be grouped into structural equivalence classes based on their K_G profiles, independent of their physical domain.
- (c) **Anomaly detection:** if a system's K_G profile deviates from its established equivalence class, this indicates a structural change that may precede a fault or indicate tampering.
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5 WORKED EXAMPLES

5.1 Example 1: Intrinsic Geometric Diagnostics (Nonlinear Regime)

System: A semiconductor diode operating at $V = 0.65$ V, $I = 72$ mA, with thermal voltage $S = 0.026$ V. The coupling equation is $F(V, I, S) = I - I_0(e^{V/S} - 1) = 0$.

Step 1: Compute the partial derivatives of F at the operating point.

Step 2: Compute the bordered Hessian BH and gradient magnitude $|\text{grad}F|$.

Step 3: $K_G = \det(BH) / |\text{grad}F|^4 = 0.105$ (positive, indicating bowl geometry).

Step 4: Isotropy ratio $R = 1.000$ (one component -- current I -- enters linearly, confirming it is the linear passenger).

Step 5: $\det(H_F) = 0 \rightarrow$ Class 1 coupling.

Diagnosis: The positive K_G indicates the coupling is in a stable nonlinear regime. The isotropy ratio identifies current as the linear passenger. The Class 1 classification is consistent with the diode equation structure (I enters linearly; V and S enter nonlinearly via the exponential).

5.2 Example 2: Recursive Self-Audit

System: Landau theory of phase transitions, $F(\phi, T, h) = (T - T^c)\phi + \phi^3 - h = 0$.

Level 0: At the critical point ($\phi \rightarrow 0, T = T^c$), $K_G^0 = 1.000$. Class 1 (h is linear passenger).

Level 1: Construct the meta-constraint from the Level 0 outputs (K_G , distance from critical point τ , profile parameter b). The meta-surface $F^1(K, \tau, b) = K \cdot (4\tau^2 + b\tau + 1)^2 - 1 = 0$ has $K_G^1 = 0.155$. Class 2 (no linear passenger at the meta-level).

Class transition detected: The coupling is Class 1 at Level 0 but Class 2 at Level 1. The recursive self-audit reveals hidden complexity not visible to single-pass analysis.

Closure tuple: (1.000, 0.155, 3, approximately equal to 0).

5.3 Example 3: Centrifuge Validation

System: The same Landau system as Example 2, with variables (ϕ, T, h).

Centrifuge result: 6 role assignments tested. The natural coupling is $D = T, S = \phi, P = h$ (temperature drives the order parameter through the field). 4 of 6 configurations are degenerate. The non-degenerate configurations correspond to the physically meaningful role assignments.

5.4 Example 4: Cross-Domain Equivalence

System A: Van der Waals gas near its critical point. K_G profile computed as a function of reduced temperature.

System B: Curie-Weiss magnet near the Curie point. K_G profile computed as a function of reduced temperature.

Direct comparison: RMS mismatch = 0.35 (significant).

After harmonisation: K_G profiles match to $\text{RMS} < 10^{-4}$. The two systems are parametrically different but geometrically equivalent -- they share the same constraint surface shape despite being

different physical systems. Both exhibit $K_G \rightarrow 1$ at the critical point (a universal geometric invariant of critical phenomena).

6 SYSTEM EMBODIMENT

The methods of the invention may be implemented as extensions to the diagnostic system described in the parent application, comprising:

- (a) An **intrinsic geometry module** that computes K_G , H , and R from the coupling equation and sensor data, using the bordered Hessian computation described in Method 1.
- (b) A **recursive audit module** that applies the diagnostic to its own outputs, iterating until convergence and producing the closure tuple, as described in Method 2.
- (c) A **centrifuge module** that systematically tests all role assignments and outputs the natural coupling, degeneracy pattern, and complexity class, as described in Method 3.
- (d) A **cross-domain comparison module** that compares K_G profiles of two or more systems, performs harmonisation via quantile matching, and classifies system pairs as geometrically identical, parametrically different, or structurally different, as described in Method 4.

These modules may operate independently or in combination with the diagnostic method of the parent application. The intrinsic geometry module (a) may be used as a standalone diagnostic without requiring the transposition engine or invariant closure evaluation of the parent application.

7 SCOPE OF APPLICATION

The methods of the invention are applicable to any system that can be modelled as a three-component relationship $F(x_1, x_2, x_3) = 0$, including but not limited to:

- Electrical and electronic systems (fault diagnosis, component characterisation);
 - Mechanical and structural systems (vibration analysis, structural health monitoring);
 - Chemical and process systems (reaction monitoring, process control);
 - Thermodynamic systems (phase transition detection, critical point identification);
 - Signal processing systems (filter characterisation, frequency response analysis);
 - Financial and economic systems (risk coupling analysis, market regime detection);
 - Supply chain and logistics systems (anomaly detection, integrity monitoring);
 - Any system where the coupling between three measurable quantities can be expressed as an implicit constraint equation.
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8 ADVANTAGES OVER PRIOR ART

1. **Parameterisation independence:** The intrinsic geometric diagnostics produce identical results regardless of how system variables are measured or scaled, eliminating a source of diagnostic inconsistency present in spectral and holonomy methods.
 2. **Self-auditing capability:** The recursive self-audit detects hidden structural complexity that single-pass diagnostics miss, and provides a convergent characterisation at all levels of examination.
 3. **Configuration validation:** The centrifuge identifies the optimal diagnostic configuration from data, rather than requiring manual expert assignment of variable roles.
 4. **Cross-domain transfer:** The equivalence detection method enables diagnostic knowledge to be transferred between physically different systems that share the same coupling geometry, reducing the need for system-specific calibration.
 5. **Resolution limit detection:** The centrifuge identifies configurations at which the diagnostic geometric content vanishes, alerting the user to structural limits of the analysis.
 6. **Unified linear-nonlinear framework:** The generalised structured coupling property handles both linear and nonlinear systems without manual regime selection.
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9 INFORMAL CLAIMS

1. A method for diagnosing a system modelled as a three-component coupling $F(x_1, x_2, x_3) = 0$, comprising computing the Gaussian curvature of the constraint surface defined by said coupling equation at one or more operating points, and classifying the system state based on the computed curvature values.
2. The method of claim 1, wherein the Gaussian curvature is computed via the bordered Hessian determinant of the coupling equation divided by the fourth power of the gradient magnitude.
3. The method of claim 1, further comprising computing the isotropy ratio from the unit normal vector of the constraint surface to determine coupling symmetry.
4. The method of claim 1, further comprising computing the Gaussian curvature at a plurality of operating points to produce a curvature profile characterised by a critical value, approach exponent, concentration ratio, and profile shape.

5. A method for self-auditing a diagnostic of a system modelled as a three-component coupling, comprising: (a) computing a primary diagnostic of the system to obtain diagnostic output values; (b) identifying three quantities from said diagnostic output values as a new three-component coupling; (c) computing a secondary diagnostic of said new coupling; (d) repeating steps (b) and (c) until convergence; (e) producing a closure tuple comprising the primary diagnostic value, the secondary diagnostic value, the convergence depth, and the fixed-point value.
6. The method of claim 5, wherein convergence is guaranteed within a finite number of iterations by a dimensional cascade.
7. A method for validating the diagnostic configuration of a system with three measured quantities, comprising testing all possible assignments of said quantities to three defined roles, computing the diagnostic for each assignment, and identifying the natural coupling as the assignment producing the most stable and informative diagnostic.
8. The method of claim 7, further comprising identifying degenerate configurations and determining the coupling complexity class from the determinant of the Hessian matrix.
9. A method for detecting structural equivalence between two systems, each modelled as a three-component coupling, comprising: (a) computing a normalised curvature profile for each system; (b) computing an optimal monotonic reparameterisation between the two profiles via quantile matching; (c) computing the residual between the harmonised profiles; (d) classifying the system pair as geometrically identical, parametrically different, or structurally different based on the residual magnitude and spectral content.
10. The method of claim 9, further comprising a destructive interference step in which the normalised profiles are subtracted and the residual is analysed via Fourier decomposition to identify whether differences are in overall shape or fine structure.
11. A system implementing the method of any preceding claim, comprising a computational processor, input means for receiving system data, and output means for presenting diagnostic results.
12. A computer-readable medium containing instructions that, when executed by a processor, cause the processor to perform the method of any of claims 1 to 10.

END OF PROVISIONAL SPECIFICATION